

LLM-UMLS-PICO: A Large Language Model-Based PICO Extraction Method with UMLS Semantic Validation

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Abstract. PICO elements serve as core information for clinical evidence retrieval and automated systematic reviews. However, existing literature often disperses PICO components across lengthy abstracts, making manual annotation time-consuming and susceptible to subjective bias. Existing methods exhibit significant limitations in handling paragraph-level noise, cross-sentence entity recognition, and knowledge constraints. To address this, we propose the LLM-UMLS-PICO method. This approach combines the semantic generation and contextual modeling capabilities of large language models (LLMs), the multidimensional consistency verification of the lightweight Transformer Verifier, and concept correction driven by the UMLS knowledge base to achieve high coverage and accuracy in candidate entity extraction. The method optimizes boundary and semantic consistency through paragraph filtering, sentence segmentation, and knowledge constraints, significantly reducing noise and cross-sentence errors in complex clinical texts. Experiments on the Cochrane dataset show that the proposed LLM-UMLS-PICO method consistently outperforms the compared baselines, achieving up to 39.9% improvement in F1-score together with clear gains in both precision and recall.

Keywords: Medical Entity Recognition · PICO extraction · LLMs · UMLS Integration · Domain-specific Evaluation

1 Introduction

The extraction of structured knowledge from unstructured text is an important research direction in natural language processing (NLP), and named entity recognition (NER) is one of its core techniques. Clinical PICO (Population, Intervention, Comparison, Outcome) element extraction from biomedical literature is a typical medical-domain information extraction task, which directly supports systematic reviews, evidence screening for meta-analyses, and clinical decision support systems. As the volume of clinical trial literature continues

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M. De Marsico et al. (Eds.): ICPR 2026, LNCS 16816, pp. 514–529, 2027.
https://doi.org/10.1007/978-3-032-31666-0_34

to grow, manual annotation can no longer support efficient evidence synthesis, making automated PICO extraction a key technical requirement for the digital transformation of evidence-based medicine. To address the issue of scarce annotated data, existing studies have attempted to alleviate this limitation by constructing multi-level annotated medical corpora [1]. Although existing deep learning–based NER methods have achieved certain success in PICO extraction tasks, they still suffer from notable limitations: these models rely heavily on large-scale manually annotated corpora and exhibit degraded performance in low-resource scenarios [2], while their generalization ability remains limited when confronted with cross-domain settings, non-standardized document formats, or multilingual environments [3]. Meanwhile, the lack of semantic consistency constraints makes extraction results prone to hallucinations and misclassification, and the robustness of the models is further degraded under noisy labels and data-scarce conditions [4,5].

With the rapid advancement of large language models (LLMs), which demonstrate strong capabilities in semantic understanding and few-shot learning, a new technical paradigm has emerged for automated PICO extraction [6]. Medical knowledge bases such as the Unified Medical Language System (UMLS) can introduce domain-specific semantic priors, thereby improving the consistency and reliability of entity recognition and classification [7]. Therefore, integrating large language models (LLMs) with medical domain knowledge to construct an efficient, scalable, and semantically consistent PICO extraction method is of significant research importance and practical value [8].

Based on the above, this paper conducts an in-depth investigation with the following main contributions:

1. Proposes a UMLS-based semantic validation method for LLM-driven PICO extraction: By integrating the semantic generation and contextual understanding capabilities of LLMs into clinical text structuring tasks, combined with structured prompts and a semantically driven candidate generation strategy, the LLM-UMLS-PICO method achieves high recall recognition and structured extraction of the four PICO elements. This provides an efficient and interpretable new paradigm for evidence-based medical information mining.
2. A mechanism for validating semantic consistency and correcting errors based on UMLS has been designed: Through Metathesaurus concept matching (CUI) and Semantic Network–based semantic type mapping (Primary Semantic Type → PICO), combined with boundary optimization and multi-source confidence fusion, candidate entities are normalized to standard medical concepts, thereby significantly improving the semantic consistency and professional reliability of the extraction results.
3. A modular, cross-domain adaptive information extraction workflow is proposed: The two-stage process comprises "semantically driven candidate generation" and "knowledge-constrained optimization," supplemented by a lightweight Transformer Verifier for context consistency evaluation. It enables rapid deployment across diverse datasets and clinical domains with-

out additional fine-tuning, demonstrating strong generalization capabilities and methodological scalability.
4. The method's effectiveness and robustness were systematically validated across multiple datasets: Experiments conducted on the standard benchmark dataset EBM-NLP mod and the real-world clinical review dataset Cochrane demonstrated that our approach achieves significant improvements in F1-score (up to 39.9%) compared to traditional models. It also maintains stable performance under cross-context and diverse terminology conditions, fully validating the method's effectiveness and generalizability.

2 Related Work

Early PICO element identification primarily relied on rule-driven and dictionary-matching approaches, including regular expressions, syntactic pattern design, and domain terminology construction, to extract structured information from medical literature abstracts or clinical trial reports. Chabou et al. [9] combined conditional random fields with rule-based methods to mitigate extraction challenges in low-resource scenarios. While this approach improved model interpretability to some extent, its performance remains highly dependent on corpus representativeness, particularly exhibiting significant shortcomings in recall. Subsequently, Dhrangadhariya et al. [10] extended PICO to a PICO tagging system using weakly supervised methods to reduce manual annotation costs. However, the noise introduced by weakly supervised tags and the high diversity of medical entities remain significant challenges.

With the advancement of deep learning technology, sequence annotation methods based on neural networks have gradually replaced rule-based approaches to become the mainstream technical approach for PICO identification. Brockmeier et al. [11] employed recurrent neural networks for named entity recognition and integrated PICO features into the relevance classification task, effectively enhancing the efficiency of systematic review screening. However, the model remains reliant on substantial annotated data and exhibits limited cross-domain generalization capabilities. To further enhance extraction accuracy, Zhang et al. [12] proposed a stepwise medical NER framework. By first performing sentence-level classification followed by entity recognition, they achieved finer-grained PICO extraction. However, its overall structure is relatively complex. To address the issue of insufficient annotated data, Zhao et al. [13] introduced large language models for data augmentation and multi-scale feature extraction, which partially mitigated noise interference. However, the reliability of their generalization performance—which relies on ChatGPT for data generation—remains to be further validated.

Building upon this foundation, pre-trained language models based on Transformers provide more context-aware semantic representations for medical text understanding through self-supervised learning on large-scale unlabeled corpora. Lee et al. [14] proposed BioBERT, which achieves significant performance improvements in tasks such as named entity recognition, relation extraction,

and question-answering through additional pre-training on biomedical domain corpora. This effectively addresses the issue of general models' insufficient adaptability in specialized domains. Subsequently, relevant studies systematically analyzed the applicability of different model architectures and training strategies in medical NLP tasks, and explored the potential applications of generative models and large-scale clinical language models in scenarios such as electronic medical record understanding and multi-task learning [15–17].

In recent years, large language models (LLMs) have further expanded the technical boundaries of PICO identification through their powerful semantic understanding and generative capabilities. Through carefully designed prompts, LLMs can accomplish information extraction tasks under zero-shot or few-shot conditions, demonstrating outstanding performance in cross-sentence reasoning, long-text modeling, and semantic normalization [18–21]. Previous studies have applied LLMs to real-world medical knowledge extraction and automated systematic review processes, such as constructing medical knowledge graphs using LLMs to support patient retrieval and disease mapping [22], and automatically generating PICOs abstracts to enhance systematic review screening efficiency [23]. Further research indicates that LLMs demonstrate improved performance in specialized retrieval tasks as model scale increases [24], while integrating retrieval-enhanced generation or proxy architectures can further enhance the evidence reliability of responses; However, in structured PICO extraction tasks, extraction-based methods still hold advantages in terms of stability and F1-score [25].

To further enhance the accuracy and interpretability of PICO extraction results, researchers have begun incorporating medical knowledge bases to impose semantic constraints and standardize model outputs. Relevant studies indicate that knowledge embedding, concept mapping, and semantic type constraints can effectively mitigate semantic ambiguity in medical texts and significantly enhance extraction performance [26–28]. Building upon this foundation, some studies have further expanded the PICO labeling system or introduced knowledge-driven decision support frameworks to enhance information extraction capabilities in complex clinical scenarios, thereby providing greater reliability and operability for automated PICO applications [29,30].

3 Methods

This paper presents a two-stage method for extracting PICO elements, which combines the semantic understanding abilities of large language models (LLMs) with the structural constraints found in medical knowledge bases (see Fig. 1). The core idea is to first generate candidate entities with high recall using large models, then perform concept validation and semantic normalization using a structured medical knowledge system to achieving both high recall and high precision. Additionally, this method introduces a lightweight Transformer-based verifier and a confidence fusion function to enhance boundary judgment and noise filtering capabilities, thereby improving model robustness and cross-corpus transferability.

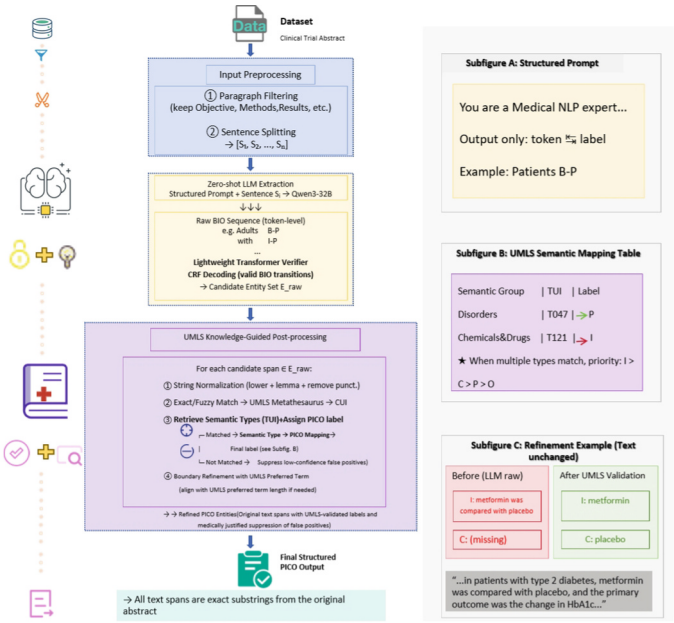


Fig. 1. LLM-UMLS-PICO method framework

3.1 Semantic-Driven Candidate Extraction Based on Large Language Models

The objective at this stage is not to directly obtain final PICO annotations but to leverage the semantic reasoning capabilities of LLMs to maximize medical concept coverage and generate a high-recall entity candidate set.

Step 1: Large Models as Semantic Generators

Traditional BioNER models are predominantly based on sequence tagging. Such models rely on limited corpora and static parameters, exhibiting weak capabilities in handling long-tail entities, implicit clinical descriptions, and cross-sentence dependencies within real medical contexts. In contrast, high-parameter LLMs possess capabilities for cross-referential reasoning, contextual integration, and understanding clinical expressions. Here, X denotes the input token sequence and Y denotes the corresponding BIO label sequence over the P/I/C/O categories. In this study, LLMs are treated as semantic generators according to formula (2). Their output is not a final sequence label but a candidate entity set C(X) encompassing potential P/I/C/O elements.

$$P(Y|X) = f_\theta(X) \tag{1}$$

$$C(X) = \mathrm{LLM}_\theta(X, \mathrm{Instruction}) \tag{2}$$

Step 2: Structured Prompt Design

The semantic advantages of LLMs often come with the risk of random generation, and without format constraints, outputs may be unparsable. To ensure results are usable in downstream pipelines, this study employs "structured prompts" to impose hard constraints on generation: Definition and examples of the four PICO elements; Clear task instructions (named entity recognition, NER); Fixed output format requirements (BIO annotation format, one-to-one correspondence output, JSON-parsable structure); Prohibition of generating explanatory content or unstructured text. Under these constraints, the generation distribution becomes $_\theta(y_{1,n} \mid x_{1,n}, z)$. Here, z denotes the structured prompt including the task definition, label schema, and output format. The output space is compressed into a finite set of BIO combinations y_{BIO}.

$$y_{BIO} = \{(B, I, O)^n\} \tag{3}$$

Here, B, I, and O denote the beginning, inside, and outside tags in the BIO scheme, respectively.

Step 3: Lightweight Transformer Verifier

Although LLMs demonstrate strong performance in cross-semantic extraction, their boundary consistency and entity type determination remain relatively weak. Therefore, a lightweight Transformer module is introduced as a candidate verifier (Transformer Verifier). For each candidate entity e, its contextual span is extracted. Based on formula (4), a discriminative model generates semantic embeddings, where V_ϕ is a 4–8–layer encoder, e is the candidate entity, and X is the contextual text sequence containing the entity. The Verifier outputs a continuous discrimination score $s_{\text{ver}} \in \mathbb{R}$, determining whether the candidate should be retained. Figure 2 illustrates the working mechanism of the Transformer Verifier.

$$h_e = V_\phi(e, X) \tag{4}$$

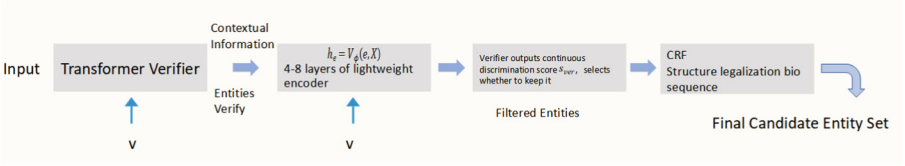


Fig. 2. Transformer Verifier utility diagram

Step 4: Sequence Structure Legitimization

Even after filtering through the Verifier, LLMs' BIO sequences often remain in an illegal structural state, such as B-P → O → I-P.

To ensure the validity of label paths, CRF is introduced as a structural constraint based on formula (5).

$$Y = \arg\max \left(\sum_{i=1}^{n} u_i(y_i) + \sum_{i=1}^{n-1} T_{y_i, y_{i+1}} \right) \tag{5}$$

Here, u_i represents token-level logits from LLMs; T denotes the label transfer matrix.

Step 5: Output of Candidate Generation Stage

This stage outputs a high-recall candidate entity set $E = \{e_1, \ldots, e_m\}$. While this set offers broad semantic coverage and strong entity capture capabilities, it still exhibits three typical shortcomings: entity boundary shifts, ambiguous semantic types, and domain/general confusion. Therefore, it relies on the UMLS medical knowledge base in the next stage for standardization and a posteriori semantic correction.

3.2 Semantic Mapping and Entity Normalization Post-processing Based on the UMLS Medical Knowledge Base

UMLS is structured and verifiable. This study employs knowledge bases as semantic discriminators, mapping candidate entities to standardized medical concepts. Semantic correction using the UMLS standard structure.

Step 1: Dual matching mechanism for lexical and semantic levels

Candidate entities undergo string normalization. Subsequently, they undergo dual matching against the UMLS concept repository $U = \{c_i\}$. Matching is performed by: 1. Lexical-level normalized set intersection matching according to formula (6) to obtain $S_{\text{lex}}(e, c)$; 2. Semantic vector matching to obtain $S(e, c)$; 3. Medical semantic embedding fusion scoring using formula (7). The optimal concept is ultimately selected:

$$S_{\text{lex}}(e,c) = \frac{|\text{Tokens}(e) \cap \text{Tokens}(c)|}{|\text{Tokens}(e) \cup \text{Tokens}(c)|}, S(e,c) \quad = \alpha S_{\text{lex}}(e,c) + (1-\alpha)\frac{\cos(v_e, v_c) + 1}{2} \tag{6}$$

$$c^* = \arg\max S(e,c) \tag{7}$$

Step 2: Concept Mapping from UMLS Semantic Type to PICO

After successfully matching to a UMLS concept c^*, perform deterministic mapping based on its primary semantic type, such as: Disease/Syndrome → P; Procedure/Drug → I; Thus, the mapping function is formally defined as formula (8). If an entity matches multiple semantic types, the highest priority type is selected according to the hierarchy: I → C → P → O. As shown in Fig. 3, combined with Subfigure B: UMLS Semantic Mapping Table from Fig. 1 (LLM-UMLS-PICO method framework), the role of UMLS Semantic Type becomes evident.

$$\Gamma : \text{SemType} \to \{P, I, C, O\}, \quad y_e = \Gamma(c^*) \tag{8}$$

Step 3: Multi-source Belief Fusion and Boundary Optimization

The final entity is determined by multi-source signal fusion. Here, pllm is obtained by averaging token-level BIO probabilities within the candidate span, s_{ver} is the verifier confidence score, and $S(e, c^*)$ denotes the UMLS matching strength. The final result is derived from Eq. (9).

$$P_{\text{final}}(e) = \sigma\left(w_1 \log \frac{p_{\text{llm}}}{1-p_{\text{llm}}} + w_2 \log \frac{s_{\text{ver}}}{1-s_{\text{ver}}} + w_3 \log \frac{S(e,c^*)}{1-S(e,c^*)} + b\right) \text{ if } P_{\text{final}}(e) < \tau_{\text{drop}} \text{ then drop} \tag{9}$$

where σ denotes the sigmoid activation function. The thresholds are selected on the validation split and kept fixed during testing. The final decision is made based on the following threshold rules: $P_{\text{final}}(e) < \tau_{\text{drop}}$: False positive removal (discard entity)$P_{\text{final}}(e) \in [\tau_{\text{drop}}, \tau_{\text{accept}})$: Weak entity retention$P_{\text{final}}(e) \geq \tau_{\text{accept}}$: Output final entity.

Candidate entities undergo string normalization using the UMLS Metathesaurus terminology dictionary, followed by exact and partial matching. For entities with successful matches but boundary anomalies, boundary refinement is performed based on UMLS standard surface form lexical length and contextual alignment rules to correct common large model issues like truncation or over-inclusion. Candidate entities failing to match UMLS concepts are treated as potential false positives and are either discarded or relabeled based on contextual confidence thresholds. Following this processing, the final output PICO entities demonstrate significantly enhanced semantic coherence and boundary accuracy, thereby improving the model's overall precision and F1-score.

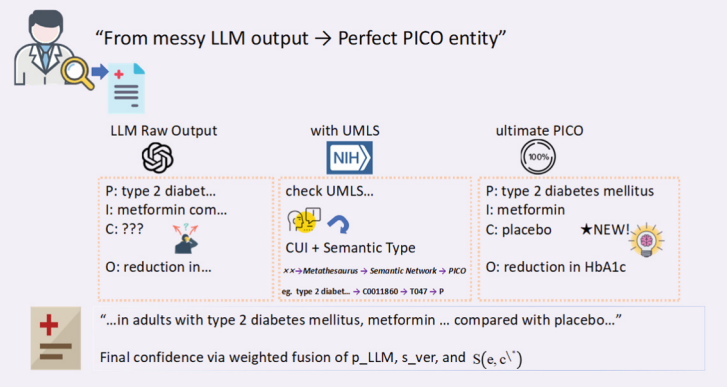


Fig. 3. Mapping Rules from Primary Semantic Types in the UMLS Semantic Network to PICO Elements

4 Experiment

4.1 Experimental Setup

Dataset and Processing. The experiments in this study are conducted on the publicly available medical text datasets EBM-NLPmod and Cochrane. EBM-NLPmod is a publicly available dataset released by Hu et al. [33], which consists of a large number of clinical trial text segments and is annotated with PICO elements according to evidence-based medicine standards. The experiments follow the five-fold cross-validation protocol provided in the original study, where each fold consists of mutually independent training, validation, and test sets

for method evaluation and baseline comparison. The Cochrane dataset is collected from the publicly available Cochrane Library and consists of titles and abstracts of clinical systematic reviews. To ensure experimental controllability, 1,000 records were randomly sampled for the experiments. Among them, 300 records were manually verified to construct a high-quality gold-standard set for validation and testing, while the remaining samples were used for model training. The training, validation, and test sets were split at a ratio of 7:1:2. Before the experiments, the raw corpora were uniformly preprocessed through text cleaning and irrelevant content filtering to reduce noise and improve the accuracy of PICO element extraction.

Baseline Models and Methods. To comprehensively validate the effectiveness and generalizability of the proposed method, comparative experiments were designed under identical data partitioning and evaluation criteria:

(1) Domain-Specific Pre-trained Models

We select two representative biomedical domain–pretrained models as supervised baselines. BioM-ALBERT [17] is built upon the ALBERT architecture and is domain-pretrained on PubMed Abstracts and PMC Full-text corpora. PubMedBERT [27] is pre-trained entirely on biomedical corpora and places greater emphasis on semantic modeling within the medical domain. Two models are fine-tuned in a supervised manner to perform sentence-level named entity recognition for PICO elements, using the BIO tagging scheme to output four PICO categories, and are representative of mainstream domain-specific pre-trained approaches.

(2) High-Recall Candidate Generation Based on Large Language Models

This paper employs LLMs for zero-shot PICO extraction experiments. The model requires no additional training, achieving word-by-word BIO annotation solely through a carefully designed structured prompt. The prompt template incorporates simplified medical abstract examples to aid the model in understanding entity boundaries and label semantics.

(3) Semantic Mapping with UMLS and Entity Normalization Post-processing

Building upon Qwen_prompt, the proposed Qwen_prompt+UMLS method further incorporates the Unified Medical Language System (UMLS) knowledge base for semantic constraint and post-processing. This method validates and corrects candidate entities through UMLS concept matching and semantic type mapping, and further performs contextual consistency evaluation by incorporating boundary optimization and a lightweight Transformer-based verifier. The final extraction results are determined via multi-source confidence fusion.

Evaluation Metrics Description. This paper conducts experiments under identical training/validation/test splits across different task scenarios. Two distinct evaluation metric systems are designed:

(1) On the EBM-NLPmod dataset, a consistent evaluation script calculates Precision, Recall, and F1-score under two granularity levels: Exact Match and Relaxed Match. Exact Match requires the model's predicted entity boundaries to perfectly align with the ground truth categories, measuring the precision of entity recognition. Relaxed Match considers a match if the entity partially overlaps with the text, evaluating the model's semantic capture capability.

(2) On the Cochrane dataset, since data originates from both large-model extraction and human verification, boundary-level matching criteria are no longer distinguished. Therefore, evaluation relies solely on entity-level precision, recall, and F1-score. Among these, the F1-score serves as the primary metric, comprehensively measuring the accuracy and coverage of the model's extraction. Additional LLMs tests on a Cochrane subset show similar trends, and pretraining data overlap is unlikely as Cochrane is absent from standard corpora.

4.2 Results

EBM-NLPmod Dataset Experimental Results. EBM-NLPmod serves as the standard benchmark dataset for structured extraction in evidence-based medicine. It is randomly partitioned into five test subsets: test1–test5. The experiments selected BioM-Albert and PubMedBERT as baseline models for comparison under identical data partitioning and evaluation metrics. The experimental results are shown in Fig. 4, showing the exact F1 and relaxed F1 performance across the test subsets.

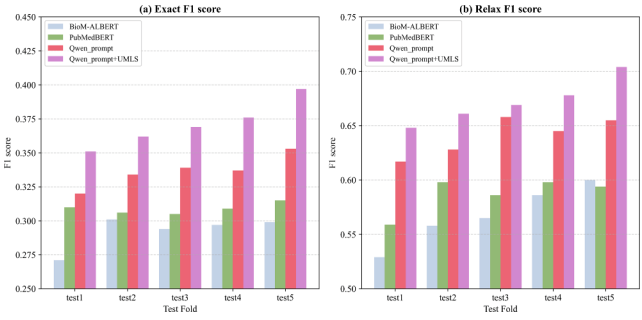


Fig. 4. Exact and relaxed F1-score trends across five test subsets of the EBM-NLPmod dataset for different models

As shown in Fig. 4, the F1-score of Qwen_prompt+UMLS shows consistently better performance across different test subsets (test1–test5), demonstrating strong cross-difficulty generalization capabilities. Compared to the baseline method, Qwen_prompt+UMLS achieved consistent improvements in both precision and recall. The Exact F1-score increased to 0.371, while the Relaxed F1-score rose to 0.672, demonstrating significant performance advantages. The results demonstrate that combining the semantic extraction capabilities of large

language models with UMLS domain knowledge constraints can effectively enhance the boundary identification accuracy and semantic consistency of PICO elements.

Cochrane Dataset Experimental Results. On the Cochrane dataset, Biom-Albert, PubMedBERT, Qwen_prompt, and Qwen_prompt + UMLS were designed for comparison. All models were evaluated under identical training and validation splits, with precision, recall, and F1-score reported for each PICO element. The overall results are shown in Table 1.

Table 1. Detailed performance metrics for each model across PICO elements

	P			I			C			O		
	F1	Rec.	Prec.	F1	Rec.	Prec.	F1	Rec.	Prec.	F1	Rec.	Prec.
Biom-Albert	0.270	0.268	0.275	0.440	0.455	0.430	0.285	0.251	0.320	0.200	0.220	0.180
PubMedBERT	0.270	0.278	0.279	0.450	0.475	0.428	0.278	0.255	0.312	0.208	0.240	0.183
Qwen_prompt	0.550	0.545	0.560	0.600	0.590	0.620	0.500	0.450	0.600	0.210	0.220	0.200
Qwen_prompt +UMLS	**0.800**	**0.820**	**0.800**	**0.810**	**0.830**	**0.790**	**0.780**	**0.800**	**0.760**	**0.220**	**0.230**	**0.210**

Comprehensive performance comparison results

Model	**Prec.** (%)	**Rec.** (%)	**F1** (%)
Biom-Albert	0.30	0.30	0.30
PubMedBERT	0.30	0.31	0.31
Qwen_prompt	0.52	0.48	0.49
Qwen_prompt+UMLS	**0.68**	**0.72**	**0.70**

Overall results are shown in Table 1. Compared to the baseline model, the Qwen_prompt approach demonstrated significant improvement over traditional pre-trained models after incorporating large-model semantic prompts, with its F1-score increasing from 0.31 to 0.49. This indicates that large models possess strong advantages in understanding complex clinical statements and extracting semantic relationships. In the comprehensive Qwen_prompt+UMLS approach, incorporating UMLS semantic post-processing further elevated the F1-score to 0.70—a 0.21 improvement over the ablation version, representing a 42.9% relative increase. Precision and recall also rose by 0.16 and 0.24, respectively, demonstrating the post-processing module's substantial gains in both accuracy and recall dimensions.

The Qwen_prompt+UMLS method significantly outperformed all baseline models and ablation models (Qwen_prompt) in terms of F1-score, recall, and precision across the three core elements: P, I, and C. The F1 score for O showed only a marginal improvement, likely due to the inherent vagueness in describing outcomes within clinical texts. In such contexts, UMLS mappings exhibit lower discriminatory power compared to distinguishing between I/C elements. For instance, in identifying interventions (I), its F1-score reached 0.81—a 35%+

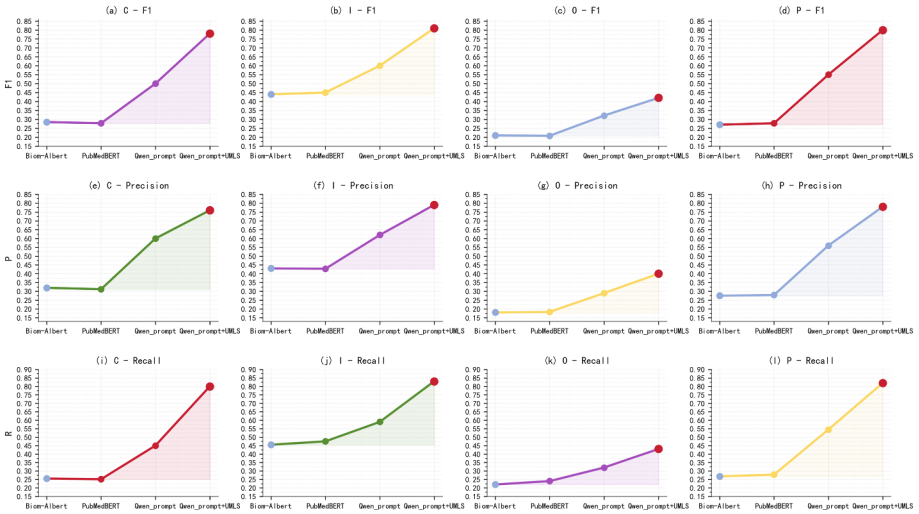


Fig. 5. Performance Comparison of PICO Elements in Cochrane Dataset

improvement over Qwen_prompt's 0.60—fully validating UMLS's critical role in normalizing complex medical intervention terminology. Similarly, the method maintains an F1-score close to 0.80 for participants (P) and controls (C), demonstrating exceptional robustness.

Figure 5 further illustrates the performance of the four methods across each PICO element. It can be observed that Qwen_prompt+UMLS achieves the highest values across all 12 metrics, with particularly significant improvement in the I element. The F1-score for I increases from approximately 0.40 to approximately 0.80, indicating that UMLS's semantic normalization capability plays a crucial role in identifying and mapping intervention and outcome-related professional entities.

In summary, Qwen_prompt+UMLS not only demonstrates outstanding performance on standardized datasets (EBM-NLPmod) but also exhibits exceptional generalization capabilities and robustness in real clinical corpora.

4.3 Ablation Studies

This paper conducts ablation studies on the Cochrane dataset. The specific design includes two configurations: the first removes the UMLS entity alignment and standardization steps, retaining only the raw extraction results (-UMLS) to measure the baseline performance of large models under plain text conditions; the second introduces an entity correction module based on the UMLS semantic knowledge base (+UMLS) to the extraction results, correcting cross-class mislabeling and boundary inconsistencies through medical concept matching and semantic category filtering mechanisms. All experiments employ a unified data

partitioning and evaluation protocol, reporting precision, recall, and F1-score for PICO elements.

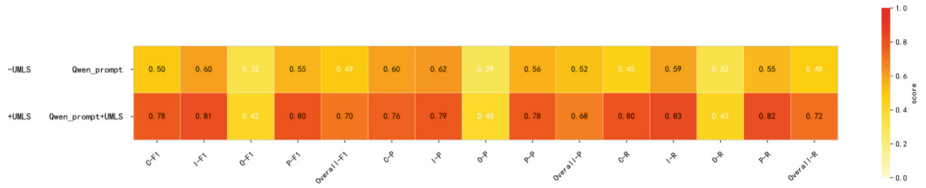


Fig. 6. Results of ablation experiments conducted on the Cochrane dataset

As shown in Fig. 6, removing the UMLS post-processing module caused the model's F1-score on the Cochrane dataset to drop from 0.70 to 0.49, primarily due to a significant decline in recall from 0.72 to 0.48. This outcome demonstrates the critical role of the UMLS module within the overall methodology. First, UMLS effectively standardizes non-standard medical terminology, demonstrating excellence in the P, I, and C elements. Second, its entity alignment mechanism based on semantic knowledge bases enhances recall while maintaining stable precision, reflecting high-fidelity error correction and conceptual constraint capabilities. Finally, the UMLS module and LLMs form a complementary closed-loop system: the former handles knowledge anchoring and semantic standardization, while the latter enables structured generation. Their synergy drives the model's evolution from open-ended generation to precise medical knowledge extraction.

Table 2. Relative Enhancement Results of the Ablation Experiment

	Precision	Recall	F1-Score	Relative Improvement
Biom-Albert	0.30	0.30	0.30	–
PubMedBERT	0.30	0.31	0.31	–
Qwen_prompt	0.52	0.48	0.49	–
Qwen_prompt+UMLS	**0.68**	**0.72**	**0.70**	+42.9%

As shown in Table 2, incorporating the UMLS knowledge base significantly enhances model performance. Extraction with UMLS achieves a 42.9% relative improvement in F1-score compared to extraction without UMLS. This improvement far exceeds the performance variation attributable to architectural differences alone, clearly demonstrating the critical value of knowledge-driven semantic alignment in medical entity extraction.

In summary, the UMLS post-processing module delivers stable F1-score gains for LLMs outputs, infusing structured extraction tasks in specialized domains with trustworthy, interpretable domain knowledge constraints. This ablation study not only validates the scientific rigor of the methodological design but also provides robust empirical evidence for the controllable improvement of LLMs in evidence-based medical scenarios.

5 Conclusions

This paper proposes the LLM-UMLS-PICO method, which leverages the sentence-level feature extraction capability of LLMs and employs post-processing via the UMLS knowledge base to achieve high-precision, semantically consistent extraction of PICO elements from clinical literature. The core contribution lies in establishing an extraction paradigm that deeply integrates LLM-based high-recall candidate generation with dual-layer UMLS knowledge constraints. Combined with a lightweight Verifier's multi-source confidence fusion mechanism, this approach significantly enhances the reliability and professionalism of extraction results, providing an efficient and interpretable new solution for applying large language models in high-reliability medical information extraction.

References

1. Nye, B., et al.: A corpus with multi-level annotations of patients, interventions and outcomes to support language processing for medical literature. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, vol. 1, pp. 197–207 (2018)
2. Jonnalagadda, S.R., et al.: Automating data extraction in systematic reviews: a systematic review. Syst. Rev. **4**(1), 78 (2015)
3. Jin, D., Szolovits, P.: Advancing PICO element detection in biomedical text via deep neural networks. Bioinformatics **36**(12), 3856–3862 (2020)
4. Tam, D., et al.: Evaluating the factual consistency of large language models through news summarization. In: Findings of the Association for Computational Linguistics: ACL 2023, pp. 5220–5255 (2023)
5. Chen, F., et al.: Semi-supervised learning from small annotated data and large unlabeled data for fine-grained PICO entity recognition, arXiv preprint arXiv:2412.19346 (2024)
6. Chen, X., Zhang, X.: Large language models streamline automated systematic review: a preliminary study, arXiv preprint arXiv:2502.15702 (2025)
7. Bodenreider, O.: The unified medical language system (UMLS): integrating biomedical terminology. Nucle. Acids Res. **32**(suppl_1), D267–D270 (2004)
8. Ghosh, M., et al.: AlpaPICO: extraction of PICO frames from clinical trial documents using LLMs. Methods **226**, 78–88 (2024)
9. Chabou, S., Iglewski, M.: Combination of conditional random field with a rule based method in the extraction of PICO elements. BMC Med. Inform. Decis. Mak. **18**(1), 128 (2018)

10. Dhrangadhariya, A., et al.: PICO to PICOS: weak supervision to extend datasets with new labels. Digital Health and Informatics Innovations for Sustainable Health Care Systems. In: Proceedings of MIE 2024, vol. 316, pp. 1775–1779 (2024)
11. Brockmeier, A.J., et al.: Improving reference prioritisation with PICO recognition. BMC Med. Inform. Decis. Mak. **19**(1), 256 (2019)
12. Zhang, T., et al.: Unlocking the power of deep PICO extraction. Step-wise medical NER identification, arXiv preprint arXiv:2005.06601 (2020)
13. Zhao, D., et al.: Few-shot biomedical NER empowered by LLMs-assisted data augmentation and multi-scale feature extraction. BioData Mining **18**(1), 28 (2025)
14. Lee, J., et al.: BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics **36**(4), 1234–1240 (2020)
15. Gu, Y., et al.: Domain-specific language model pretraining for biomedical natural language processing. ACM Trans. Comput. Healthc. (HEALTH) **3**(1), 1–23 (2021)
16. Lu, Q., et al.: ClinicalT5: a generative language model for clinical text. In: Findings of the Association for Computational Linguistics: EMNLP 2022, pp. 5436–5443 (2022)
17. Alrowili, S., Vijay-Shanker, K.: BioM-transformers: building large biomedical language models with BERT, ALBERT and ELECTRA. In: Proceedings of the 20th Workshop on Biomedical Language Processing, pp. 221–227 (2021)
18. Schmidt, L., et al.: Data mining in clinical trial text: transformers for classification and question answering tasks, arXiv preprint arXiv:2001.11268 (2020)
19. Chen, D., et al.: Extracting structure information from narrative medical reports based on LLMs. In: 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 5616–5623 (2024)
20. Mohammed, S., Fiaidhi, J.: Generative AI for evidence-based medicine: a PICO GenAI for synthesizing clinical case reports. In: ICC 2024-IEEE International Conference on Communications, pp. 1503–1508 (2024)
21. Farzi, S., et al.: Get the Best out of 1B LLMs: insights from information extraction on clinical documents. In: Proceedings of the 23rd Workshop on Biomedical Natural Language Processing, pp. 266–276 (2024)
22. Kim, E., et al.: Structured extraction of real world medical knowledge using LLMs for summarization and search. In: 2024 IEEE International Conference on Big Data (BigData), pp. 3421–3430 (2024)
23. Vallamchetla, S.K., et al.: Do it faster with PICOS: generative AI-Assisted systematic review screening. J. Biomed. Inform., 104860 (2025)
24. Demir, G.B., et al.: Enhancing systematic reviews in orthodontics: a comparative examination of GPT-3.5 and GPT-4 for generating PICO-based queries with tailored prompts and configurations. Eur. J. Orthod. **46**(2), cjae011 (2024)
25. Low, Y.S., et al.: Answering real-world clinical questions using large language model, retrieval-augmented generation, and agentic systems. Digit. Health **11**, 20552076251348850 (2025)
26. Liu, S., et al.: Sent2Span: span detection for PICO extraction in the biomedical text without span annotations, arXiv preprint arXiv:2109.02254 (2021)
27. Hu, Y., et al.: Towards precise PICO extraction from abstracts of randomized controlled trials using a section-specific learning approach. Bioinformatics **39**(9), btad542 (2023)
28. Mohammed, S., et al.: Integrating a PICO clinical questioning to the QL4POMR framework for building evidence-based clinical case reports. In: 2023 IEEE International Conference on Big Data (BigData), pp. 4940–4947 (2023)

29. Kartchner, D., et al.: TrialSieve: a comprehensive biomedical information extraction framework for PICO, meta-analysis, and drug repurposing. Bioengineering **12**(5), 486 (2025)
30. Liu, Z., et al.: PICOAS: a clinical knowledge linking model for delivering up-to-date, interrelated, and personalized decision support. In: 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 6589–6596 (2024)